

Day 9

WSD 09/10/19
Gaining H.A.

Dict Comprehensions + Sets

Recap For List $n = [1, 2, 3, 4, 5]$

squares = $[x \times x \text{ for } x \text{ in numbers}]$

names = {

"Alice" = 80,

"Bob" = 70,

"Charlie" = 95 }

{ name: len(name) for name in names }

name = len(name)

new-dic ["Alice"] = 5

" ["BOB"] = 3

Same as writing this

long_names = {

name: len(name)

for name in names

if len(name) > 3

print(long_names)

one of

{ name: len(name) for name in names }

{ value: key for key, value in student_marks.items() }

problem #4 review

Frozenset = frozen({ "a", "b", "c" })

Unordered collection of unique values

$n = \{1, 2, 3, 2, 3, 1, 2, 4, 5\}$

To check `print(2 in n)`

In list

$O(n)$ or linear

In set

$O(1)$

Use set if:

don't need duplicate values
very fast membership checking
don't give an about order

$a = \{ "Alice", "Bob", "Charlie", "David" \}$

$b = \{ "Charlie", "David", "Eve", "Frank" \}$

Union(`|`) `print(a | b)`

pipe operator

Intersection(`&`)

`print(a & b)`

ampersand

Difference(`-`)

`print(a - b)`

what exist in a that not in b

$(a, b, c, d) - (b, d, f, g)$

$= (a, c, f, g)$

Symmetric Difference(`^`)

keep element that is not common

$(a, b, c, d) \Delta (b, d, f, g)$

(a, c, f, g)

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highest = {}

name: mark

for name, mark in student_marks.items():

if mark >= 70

{

print(highest)